Spring 2025: Final Part 3

EQUAL GRADES, UNEQUAL PAY: EXAMINING INTERNAL PAY DISPARITIES IN THE DC GOVERNMENT WORKFORCE

MATH 014-01

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HOWARD UNIVERSITY

DATASET OVERVIEW

- DC Public Employee Data:
  - It includes 40,611 rows x 8 columns
  - Six categorical variables and two numerical in the original data set
  - Becoming a Howard undergraduate student means becoming a DC resident
- As I move into my career, its important for me to make informed decisions.

GOAL AND OBJECTIVE

- Iwanttobeable tohelp future DCPublic Employee job seekers to make informed decisions about where they choose to apply.
- By focusing on variables such as agency, hiring date, and grade — I can better understand the salary rate.

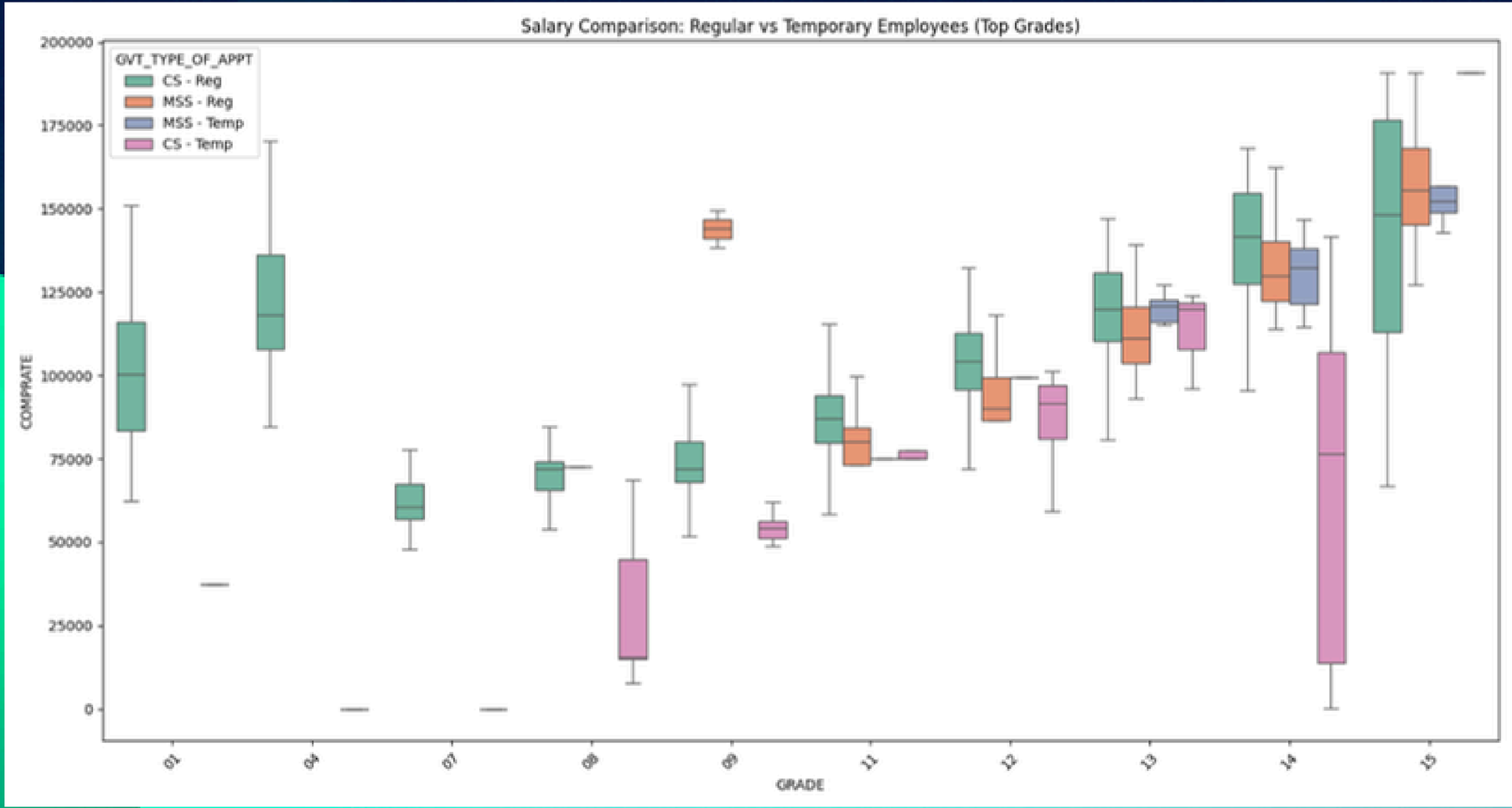
DATA CLEANING + FILTERING

- 40 missing values were in the GRADE column
- Extracted the Hiring Year from the more complex hiring date
- Removed the case sensitivity and white space
- Removed the OBJECTID column permanently
- Aggregate the Data into the most common Salary Grades, Appointment Type, and department
- Changing Grade and year from categorical to numerical

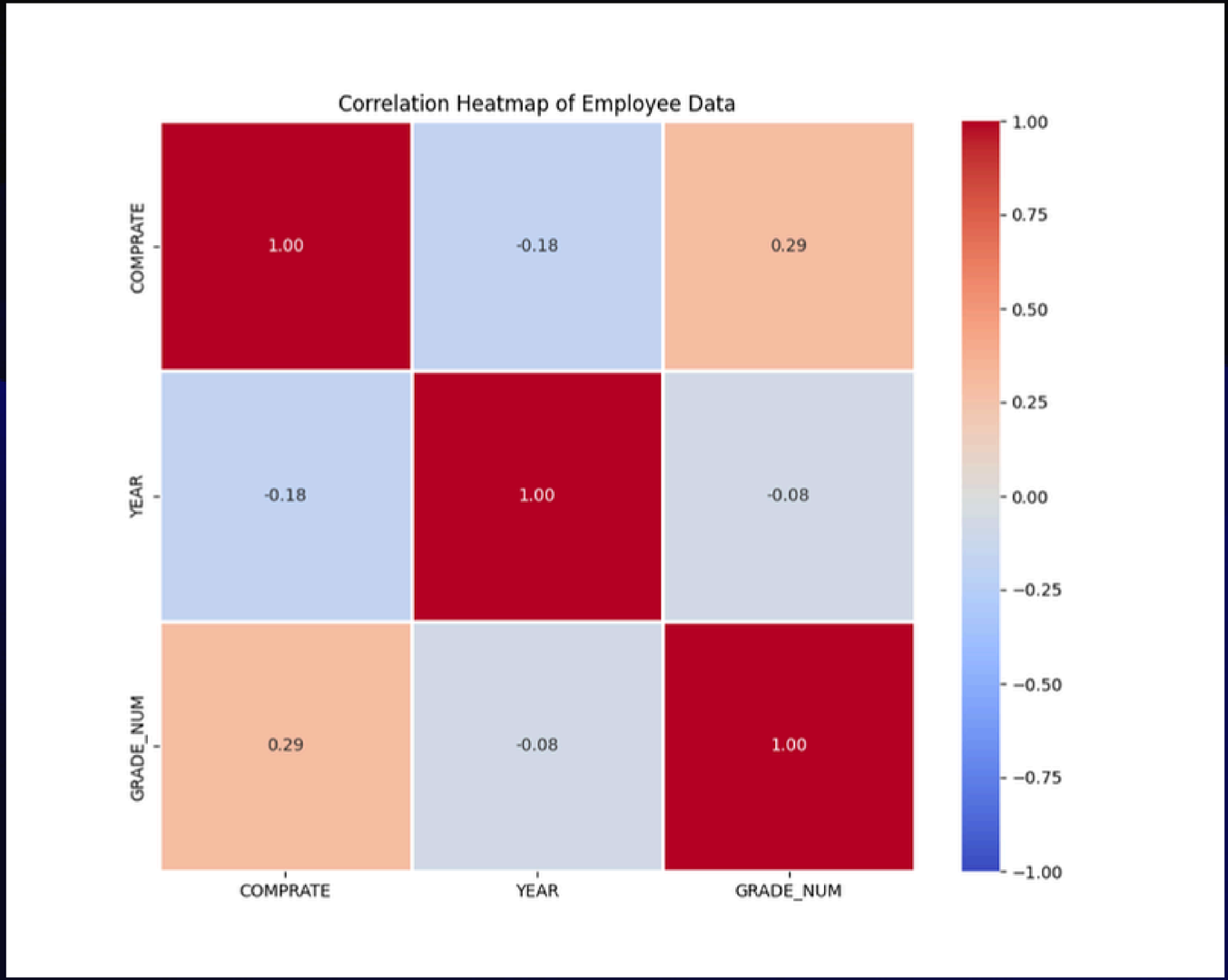
ANALYSIS QUESTIONS

- What is the correlation between hiring year and grade?
- Do employees with different appointments earn systemically different rates in the same grade?

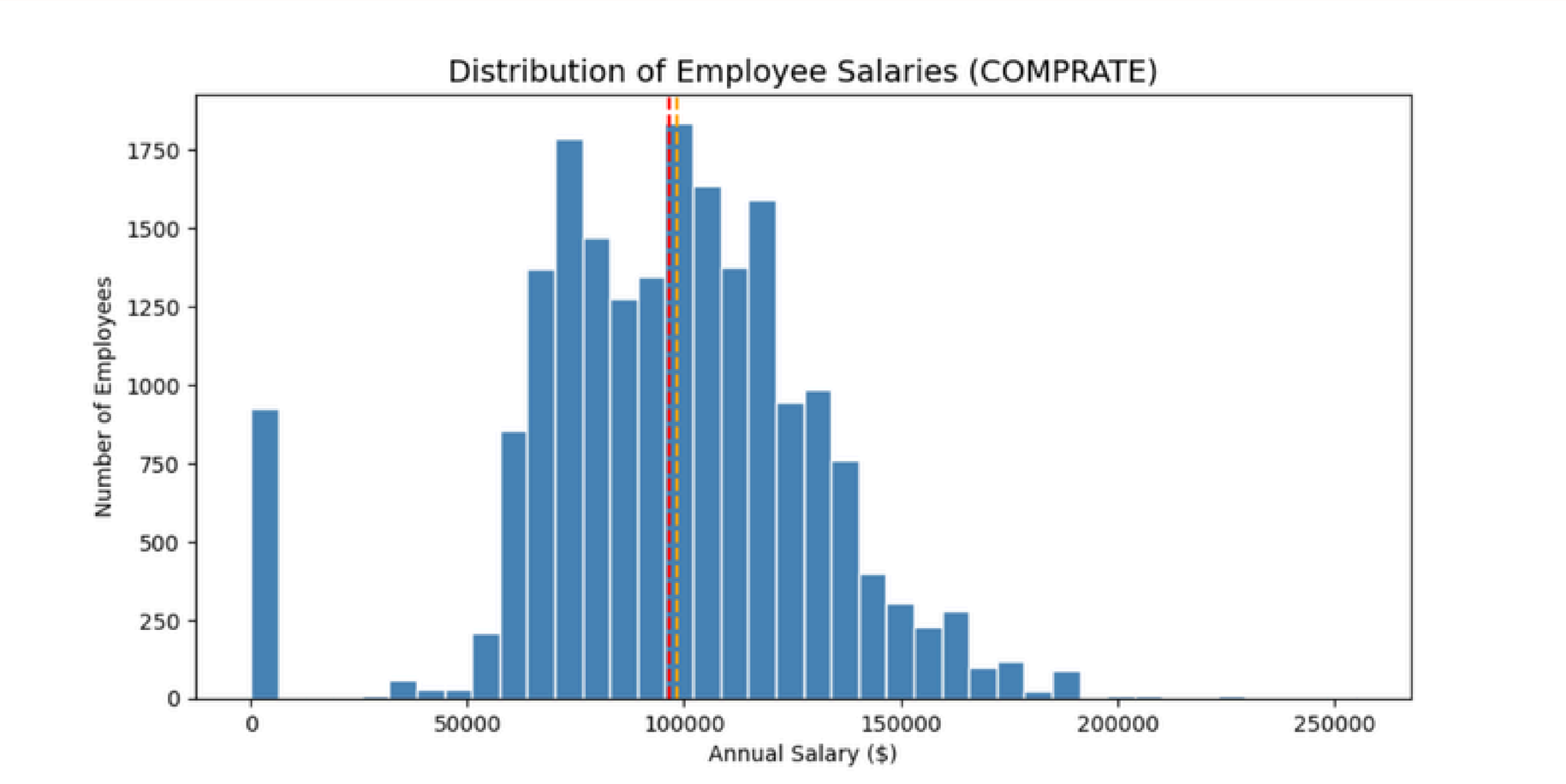
VISUALIZATION PT. 1



VISUALIZATION PT. 2



VISUALIZATION PT. 3



KEY INSIGHTS

- There is very little to no correlation between hiring year and salary grade
- Different job appointments earn systemically different salaries in the same grade
- People in CS-Temp positions can make significantly less money than any other appointment

CONCLUSION

- Due to the lack of job end or note about progression its hard to make conclusions about how jobseekers can expect progression or how long the employee stays. However this project is still valuable to explain how positions are currently paying.
- I would want to further study about how other factors such as employee race or gender may show disparity, because those variables weren't included I'd likely have to pivot to another data set
- This data set was most useful in seeing disparity across department as it showed that for departments such as DCPS they didn't make up the higher employee salary despite the wealth of data available.